

Principles of Species Distribution Modeling

Modeling the distribution of marine biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

What is species distribution modeling (SDM)?

Process of **using computer algorithms to estimate and predict the relationship between species distributions** (occurrence, abundance) **and the environmental.**

Purpose:

- Gain insights into species' environmental tolerances and habitat preferences.

- Make spatial predictions of geographic distributions across space and time.

Also known as: Environmental Niche Modelling, Habitat Suitability Modelling, ...

Why SDMs matter (marine focus)

Increasingly vital tools in marine science and resource management.

Ecology: understand the drivers and patterns of species distributions.

Conservation planning and Marine Spatial Planning: identify areas for conservation (e.g., MPA design); evaluate the potential of invasive species to settle in new areas; identify suitable areas for aquaculture.

Fisheries management: identify suitable areas of commercial species (e.g., habitat mapping).

Climate change ecology: project climate change impacts (e.g., feed IPCC reports).

Connecting niche theory to SDM practice

BAM Framework: highlights the factors (A, B, M) that determine distributions.

Niche concept: define what we are trying to model. Often an 'A'-based niche, thus capturing the abiotic drivers of distributions, and assuming or ignoring B and M.

Niche conservatism: a key assumption underpinning SDM applications (e.g., predicting future distributions under climate change).

Understanding these principles is essential for:

- Building appropriate SDMs;

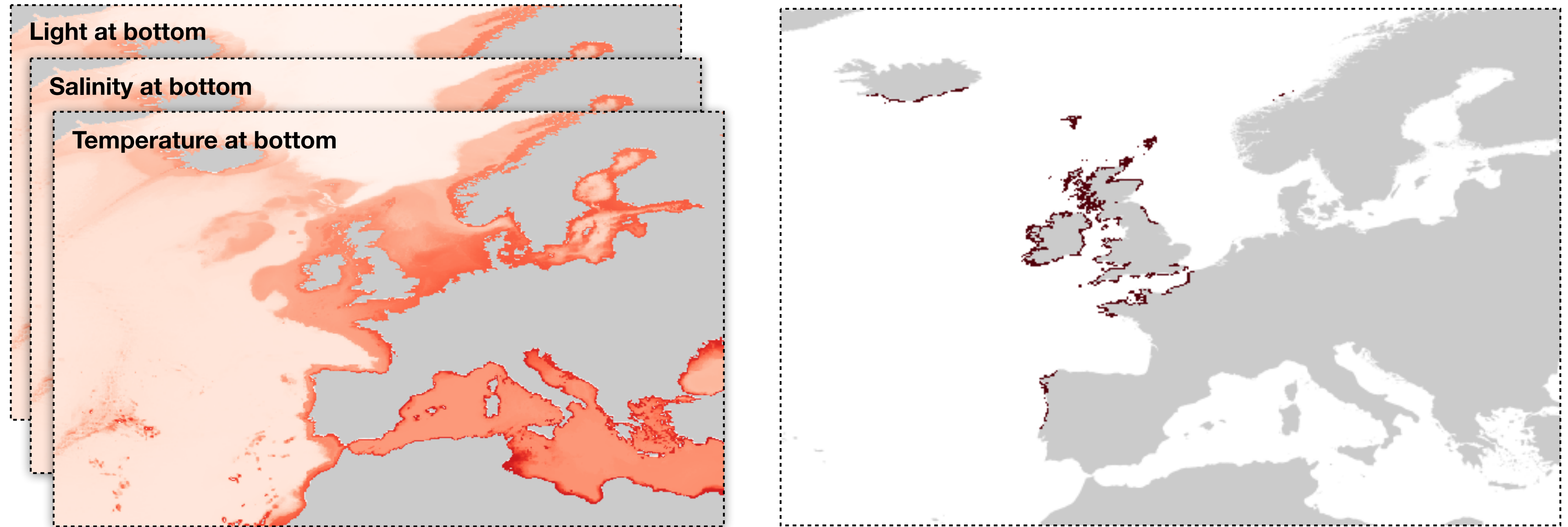
- Critically evaluating SDM outputs and limitations.

- Interpreting predictions in an ecologically meaningful way.

Main SDM approaches

Mechanistic modeling: uses **information about the physiological response of species** to environmental conditions (e.g., temperature limits for survival). Require empirical data (often from lab experiments), which may not always be available.

Correlative modeling: based on the **statistical correlation between species occurrence records** (presence-absence) **and the environment**, under the assumption that the distribution of species is an indicator of its environmental requirements.

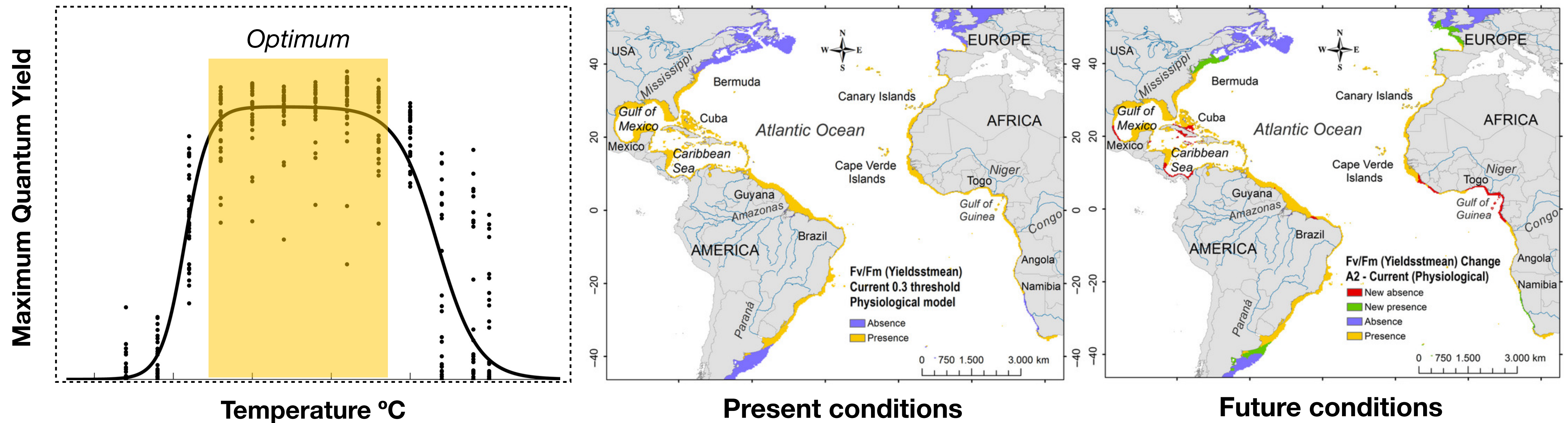


Mechanistic distribution models

Process: built by reclassifying environmental gradients with tolerance limits.

A straightforward approach to predict the distribution of species.

Presence = $[\text{Light} > 5 \text{ E.m}^2.\text{year}^{-1}] \cap [5 \leq \text{Temperature (}^{\circ}\text{C)} \leq 20.5^{\circ}\text{C}]$

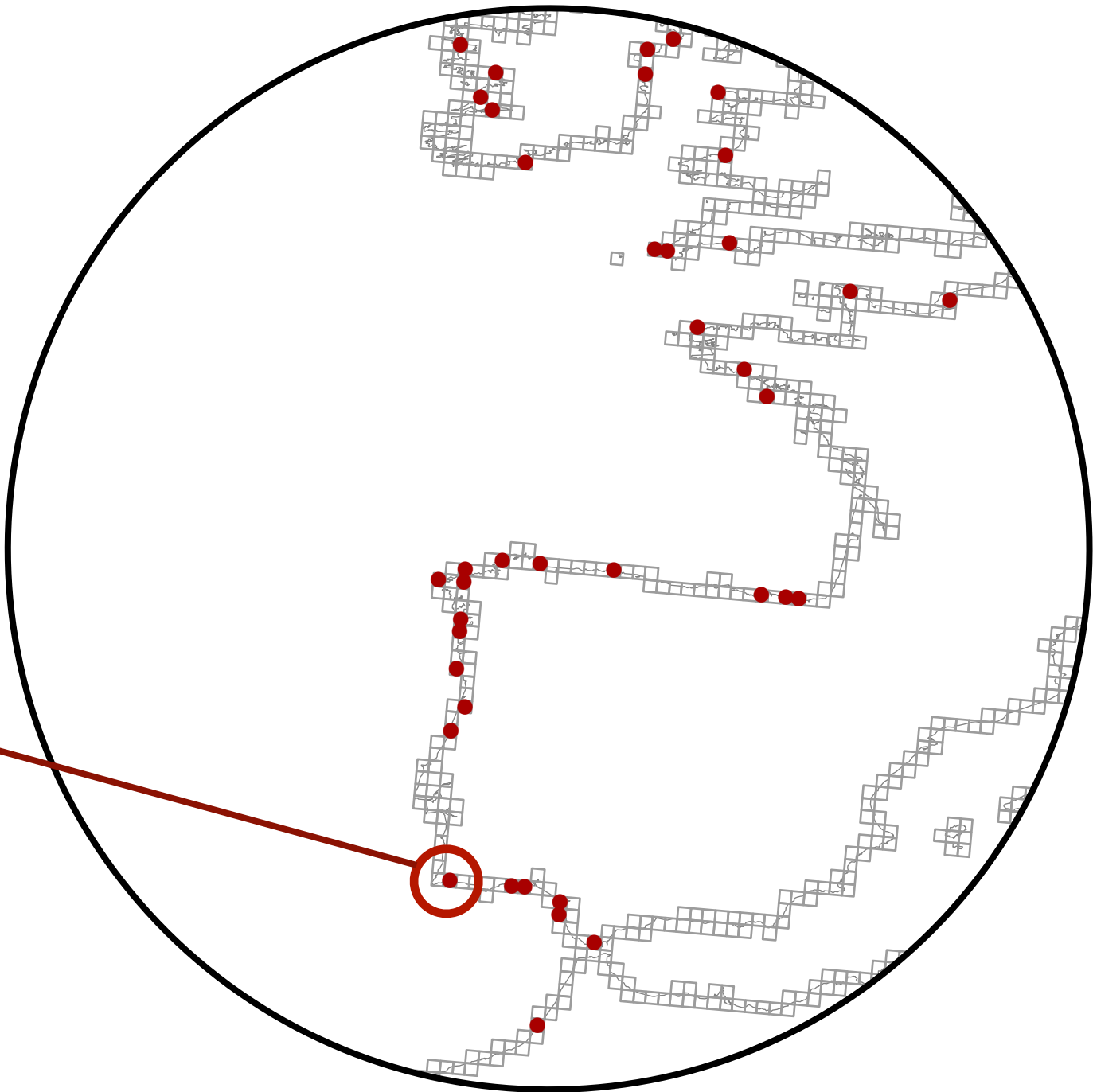


Mechanistic distribution models

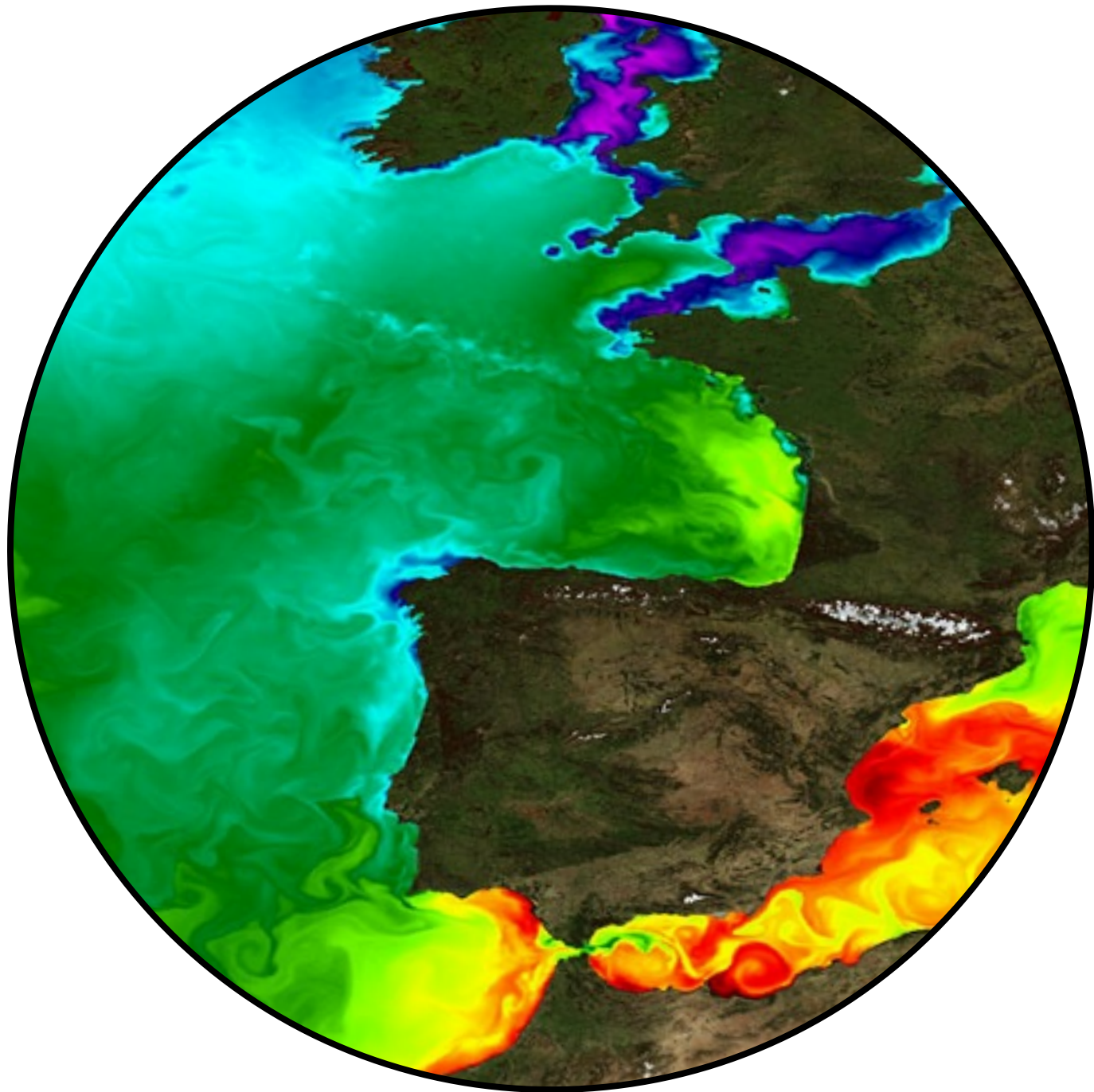
Data driven mechanistic distribution models (more realistic).

e.g., using response curves from experiments determining the tolerance of species - from cold and warm conditions (e.g., coral's symbiont performance by maximal quantum yield of photosynthesis) - the optimal conditions of the symbiont can be mapped.

	SST	Salinity	Oxygen
Site 1	21.0	35.4	210.1
Site 2	19.4	29.1	224.9
Site 3	21.5	31.1	199.8
Site 4	21.3	33.4	201.1
Site 5	14.8	34.1	184.8
(...)	(...)	(...)	(...)



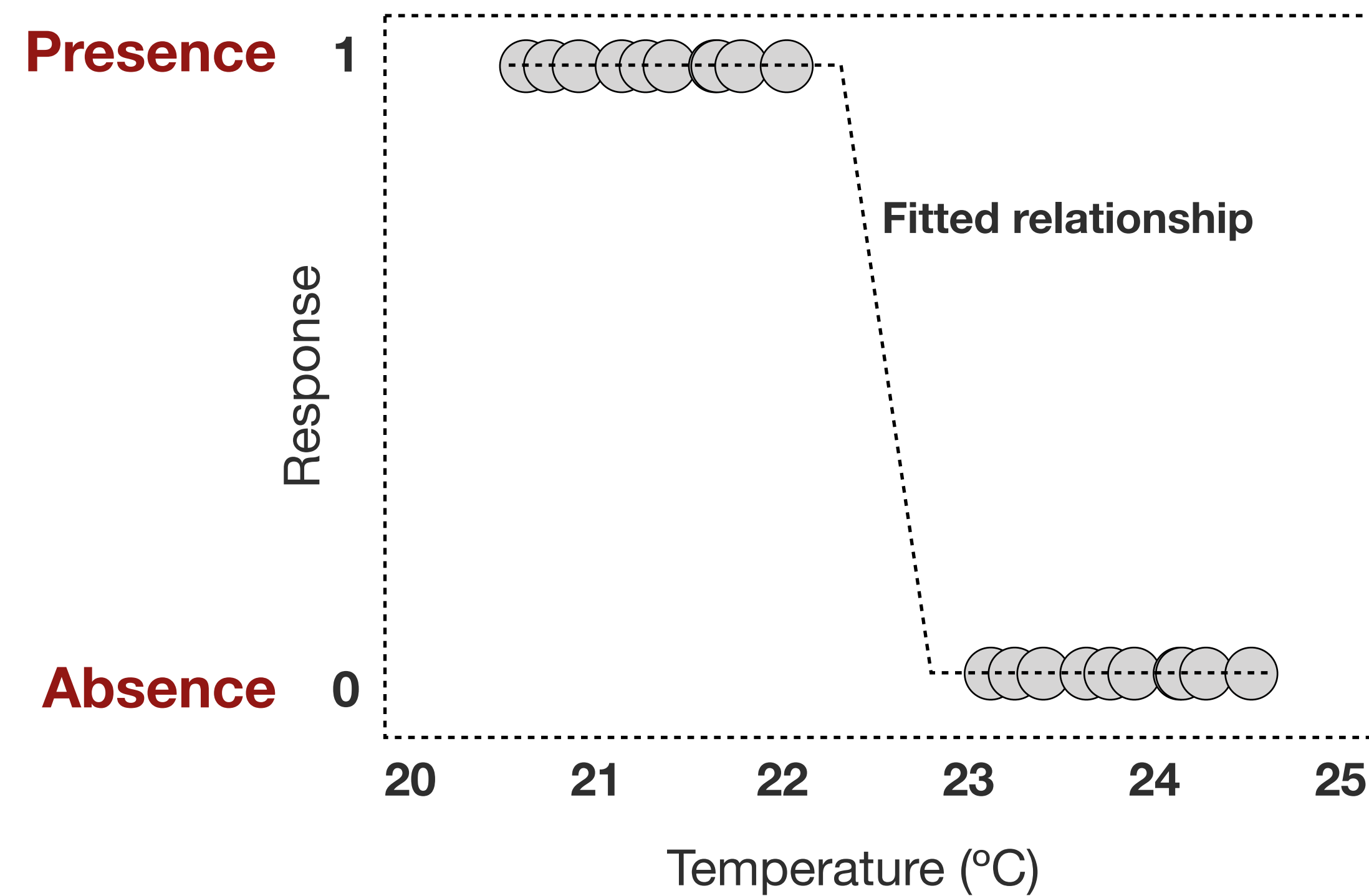
Records of species distribution



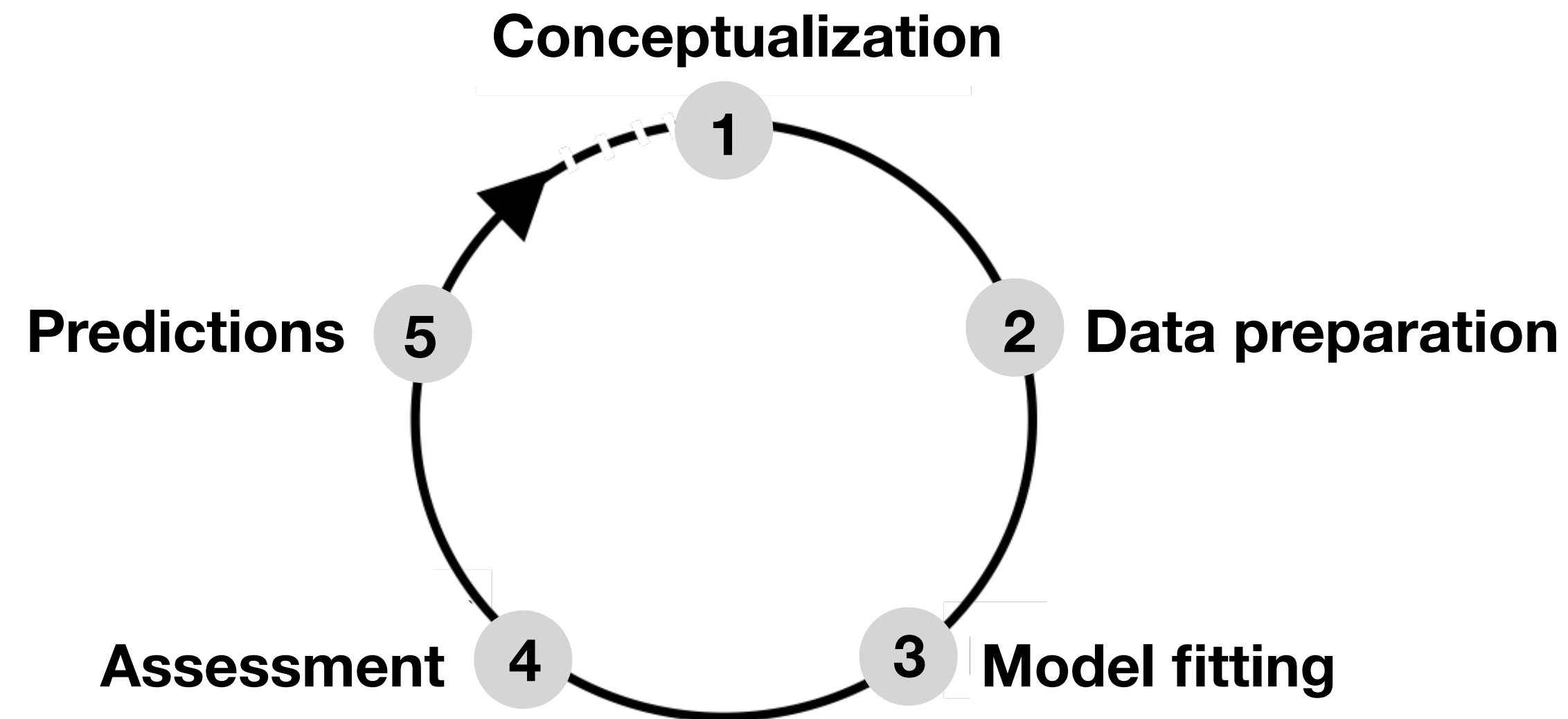
Environmental data

Correlative distribution models

Use statistical correlations between observed species occurrence records and the environmental conditions at those locations. The observed distribution reflects species' environmental requirements.



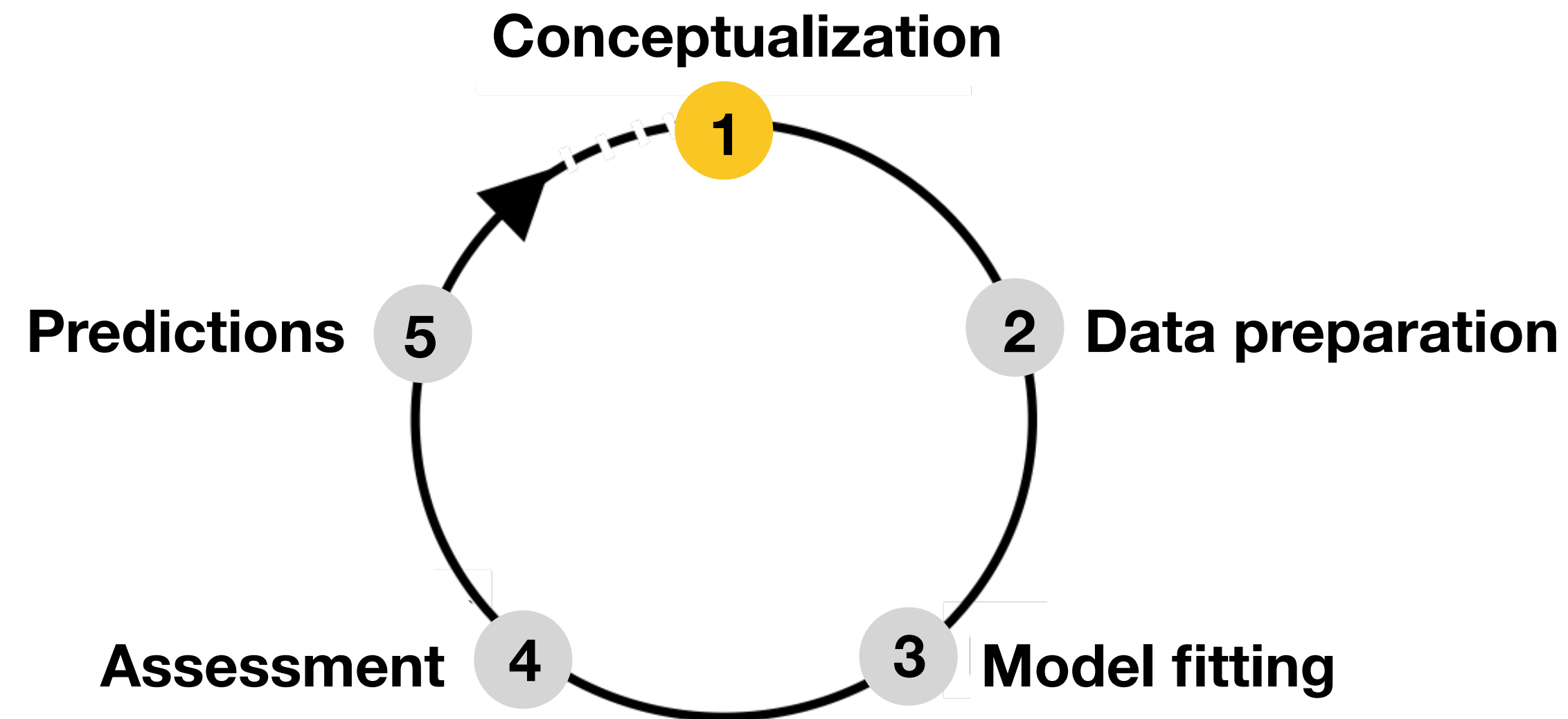
The modeling algorithm learns the effect of environmental variables on the distribution of species (training data) and fits a relationship.



Model building: an iterative loop

Expect feedback loops: building an SDM is not a linear process. It involves distinct stages, **with insights from later stages requiring revisiting earlier ones** (e.g., improve biodiversity data collection or remove surplus environmental layers).

The **goal is to build ecologically meaningful models fit for the purpose** (e.g., conservation planning and climate impact assessment).

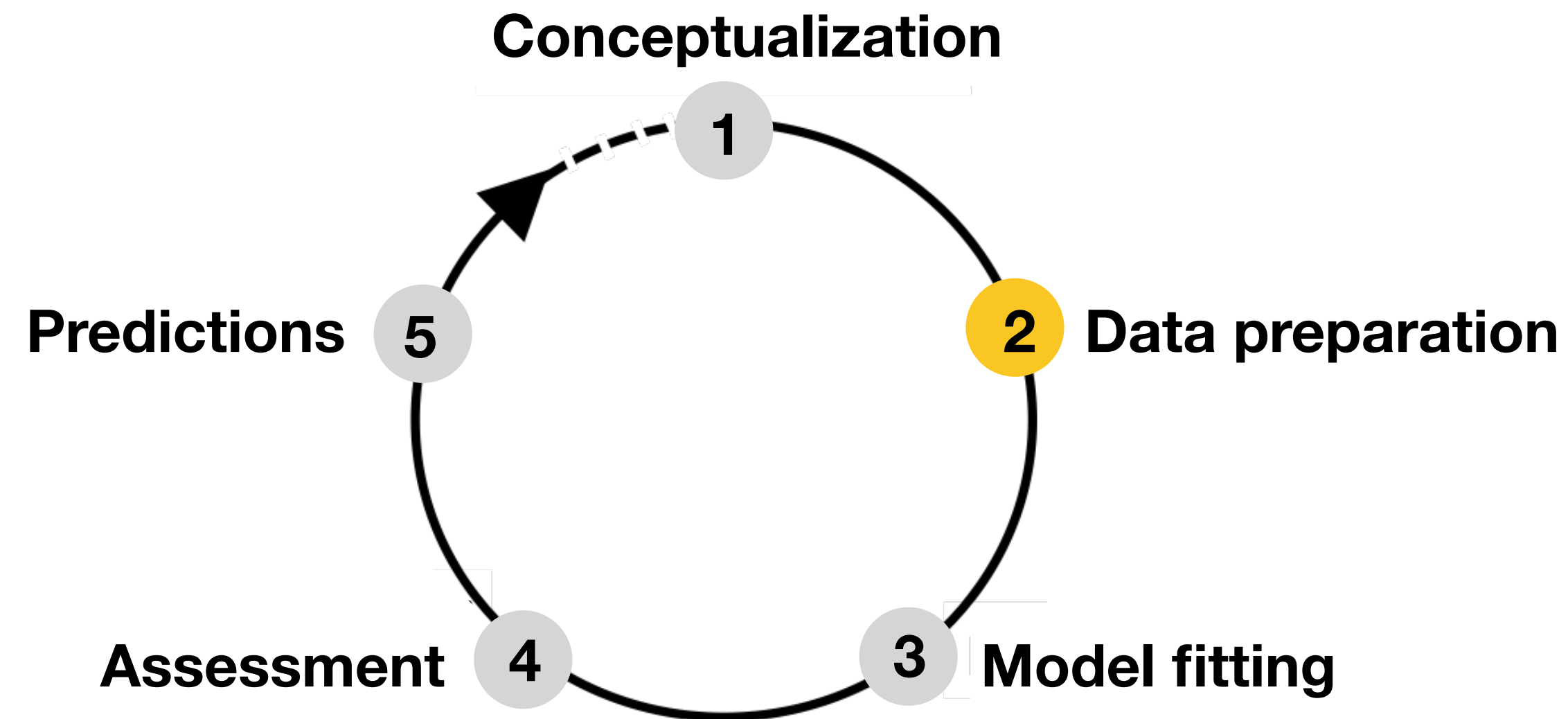


Conceptualization

Define the objective: what question is the model trying to answer? (e.g., forecast future distribution under climate change? identify invasion risk?)

Get knowledge on the species: about its basic ecology and distribution limits.

Hypothesize drivers: What variables are likely influencing its distribution?



Data preparation

Gather and process species occurrence data (e.g., check for taxonomic errors);

Select variables relevant to conceptual hypotheses.

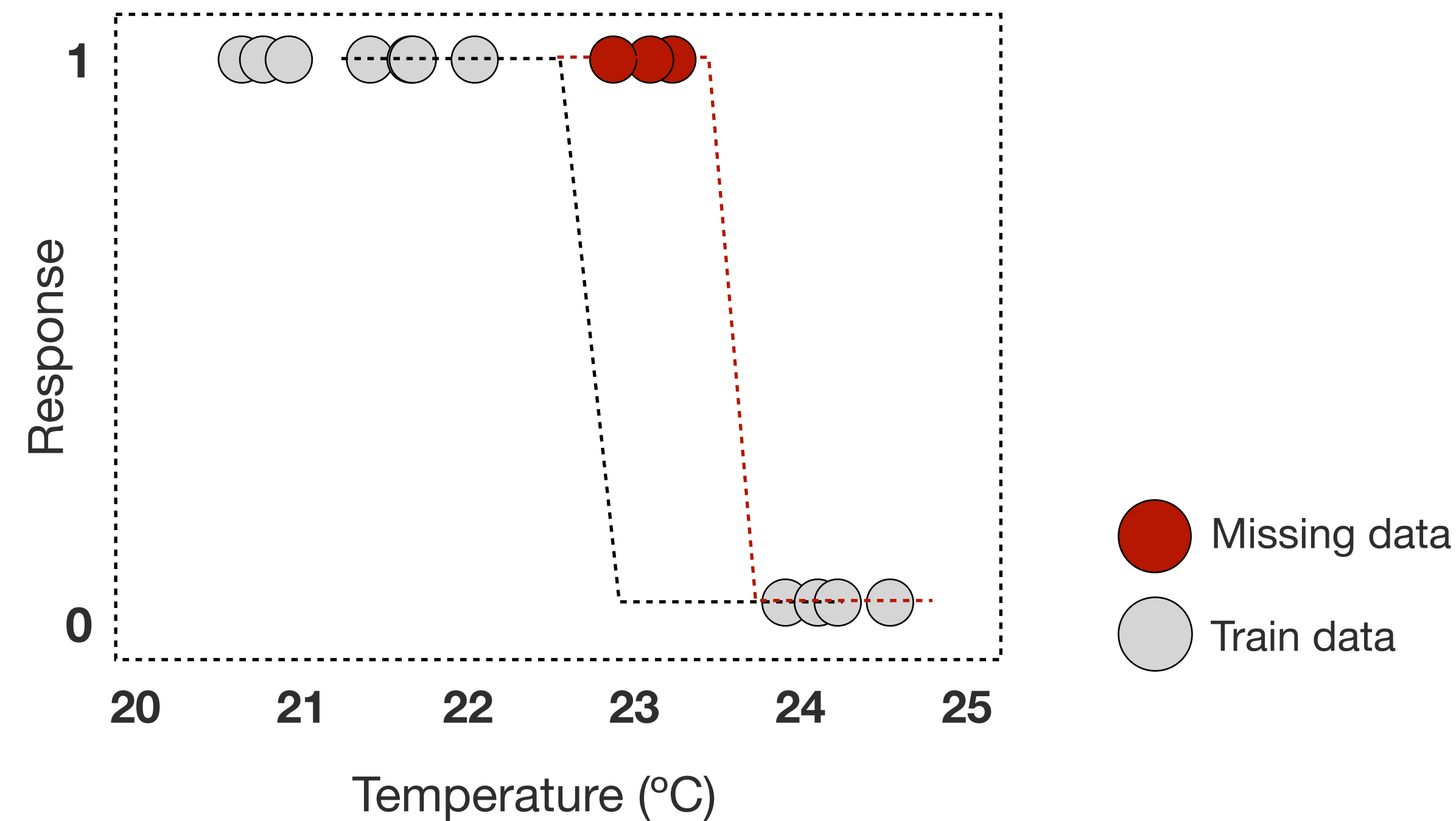
Define the study area: what geographic extent and spatial / temporal resolution are appropriate for the question and data?

How many presence records?

Sample size can be positively correlated to the performance of models.

Experiments point to **at least 50 records to estimate response functions.**

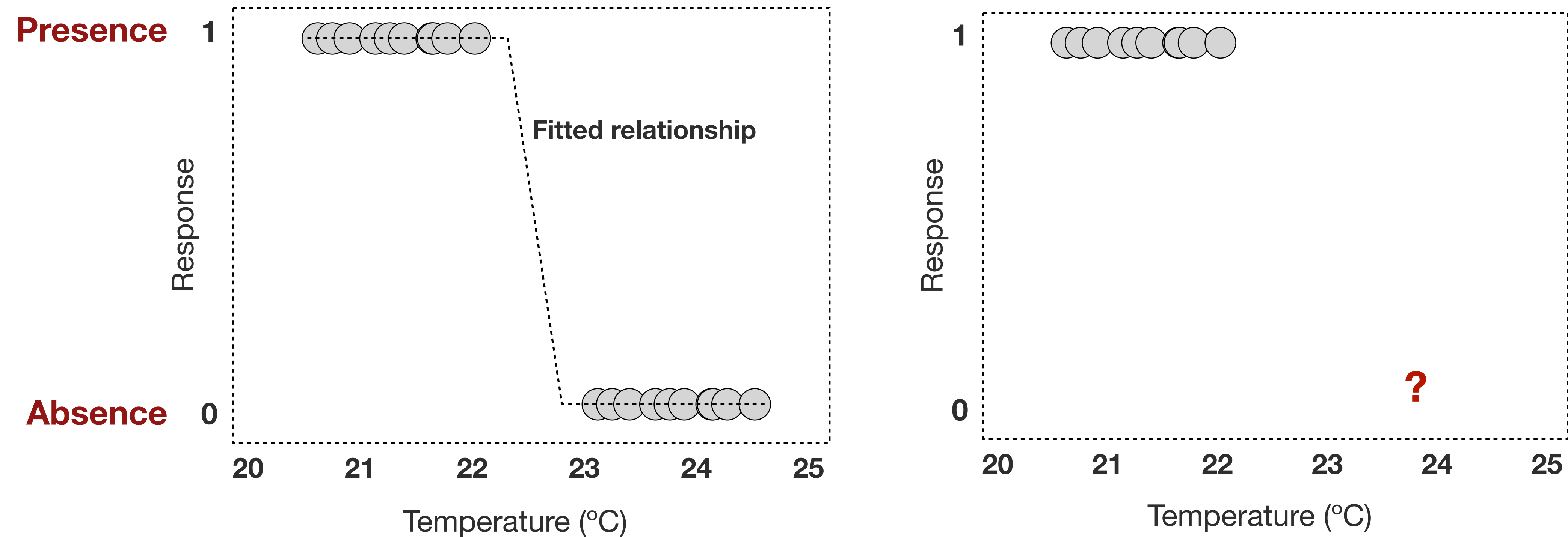
The **absolute number of observations is less important than having well-distributed observations** - cover the environmental space where the species is distributed. Thus, compiling records should be guided by knowledge on species distributions.



Niche truncation

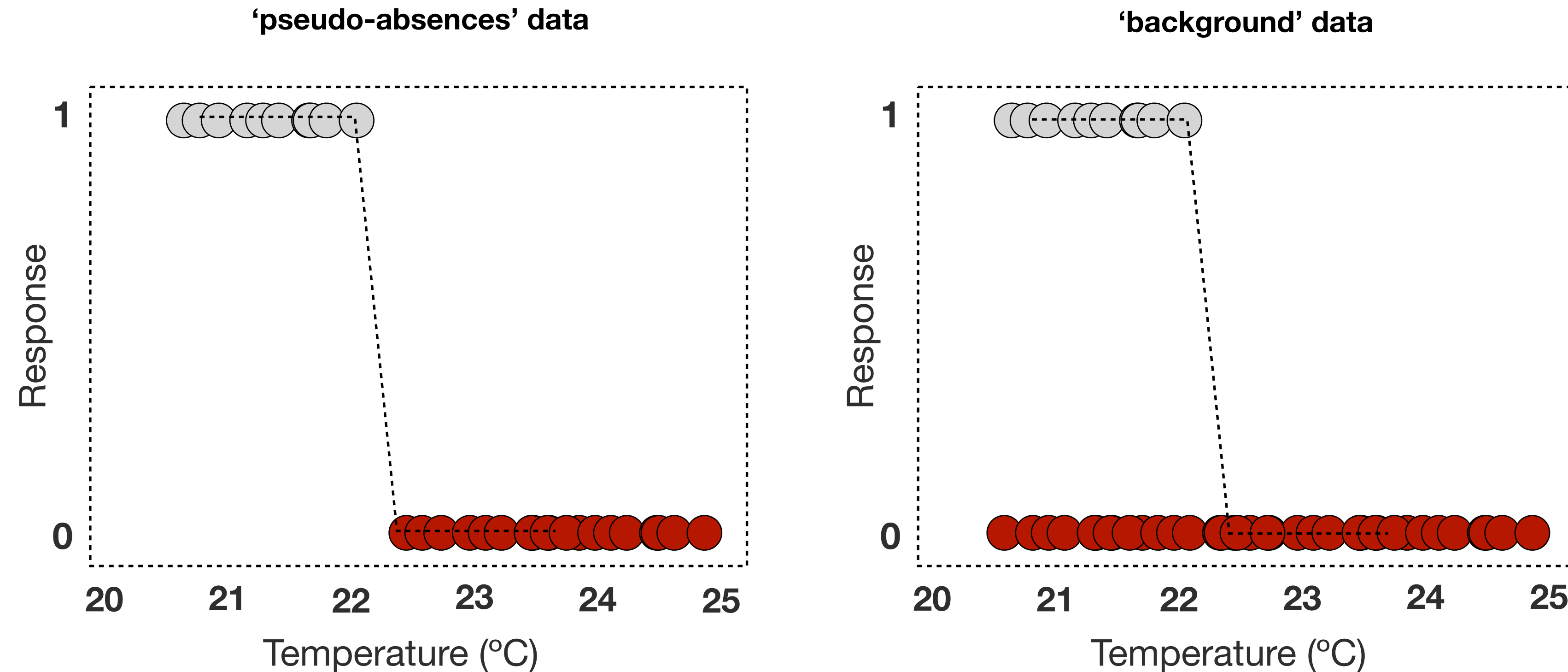
If presence records are not well distributed, the models may not capture the full width of the niche, the full span of environmental conditions which the specie is exposed to, leading to niche truncation.

The model fits the portion of the niche represented by the observed records.



Absence data

Correlative models infer **the relationship between species presence and absence records and the environment**, but **absences are often unavailable or unknown**, leading to **presence-only datasets** (common for marine species).



Strategies for presence-only modeling

Pseudo-absences: generated in sites of the study area where presences have not been recorded. Any regression algorithm can be implemented (e.g., GLM and BRT).

Background data: generated from the entire study area. Focuses on comparing the the conditions where the species occurs vs. the entire study area (e.g., MaxEnt).

How many absence records?

Pseudo-absences:

Linear models: a large number (typicality 10,000).

Advances machine learning models models (e.g., BTR and XGBoost): the same number as presence records (when less than 1000 P, 10 runs with at least 100 PA).

Background data:

Maximum entropy (MaxEnt): a large number (typicality 10,000).

Environmental variables: selection

Ecological relevance is key:

Choose variables known or hypothesized to directly influence the species' survival, growth, or reproduction.

Consider physiology and life history.

Using large datasets is an approach with stronger criticism.

Important marine variables:

Physical: temperature, salinity, bathymetry, currents velocity, sea ice conditions.

Chemical: oxygen, nutrients (e.g., nitrate and phosphate).

Biological: chlorophyll-a (proxy for productivity).

Environmental variables: selection

Marine mammals (e.g., whales, dolphins, seals):

Often driven by temperature tolerances, prey distribution, and physical barriers (e.g., sea ice).

Potential variables:

Ocean temperature: physiological tolerance limits.

Salinity: physiological tolerance limits.

Primary productivity / Chlorophyll: proxy for food web base supporting prey.

Bathymetry: depth limits for diving / foraging, habitat association.

Sea ice cover / sea ice thickness: Critical habitat for polar species (e.g., seals, polar bears).

Environmental variables: selection

Fish (e.g., Demersal Cod, Pelagic Tuna):

Temperature and oxygen are critical physiological drivers, plus factors related to food.

Potential variables:

Ocean temperature: physiological tolerance limits.

Salinity: physiological tolerance limits.

Dissolved molecular oxygen: physiological tolerance limit, especially in deeper or oxygen minimum zones.

Bathymetry: strong determinant of habitat zones (e.g., demersal species).

Primary productivity / Chlorophyll: proxy for food web base supporting prey.

(For demersal) topographic variables (e.g., slope): Proxy for habitat structure, substrate type.

Environmental variables: selection

Macroalgae & Seagrasses:

Requirements for light, nutrients, suitable substrate, temperature limits, water motion.

Potential variables:

Ocean temperature: physiological tolerance limits.

Salinity: physiological tolerance limits.

Photosynthetically Available Radiation (PAR): Essential for photosynthesis.

Diffuse attenuation: Measures light penetration depth, restricts photosynthesis.

Nitrate, Phosphate, ...: Key nutrients for growth.

Sea water velocity: Affects nutrient uptake.

Environmental variables: selection

Corals (e.g., Warm-water Reef Builders, Cold-water Corals):

Temperature sensitivity, water chemistry, light (for symbiotic warm-water corals), substrate, water flow.

Potential variables:

Ocean temperature: physiological tolerance limits.

Salinity: physiological tolerance limits.

pH: affects calcification (building skeletons).

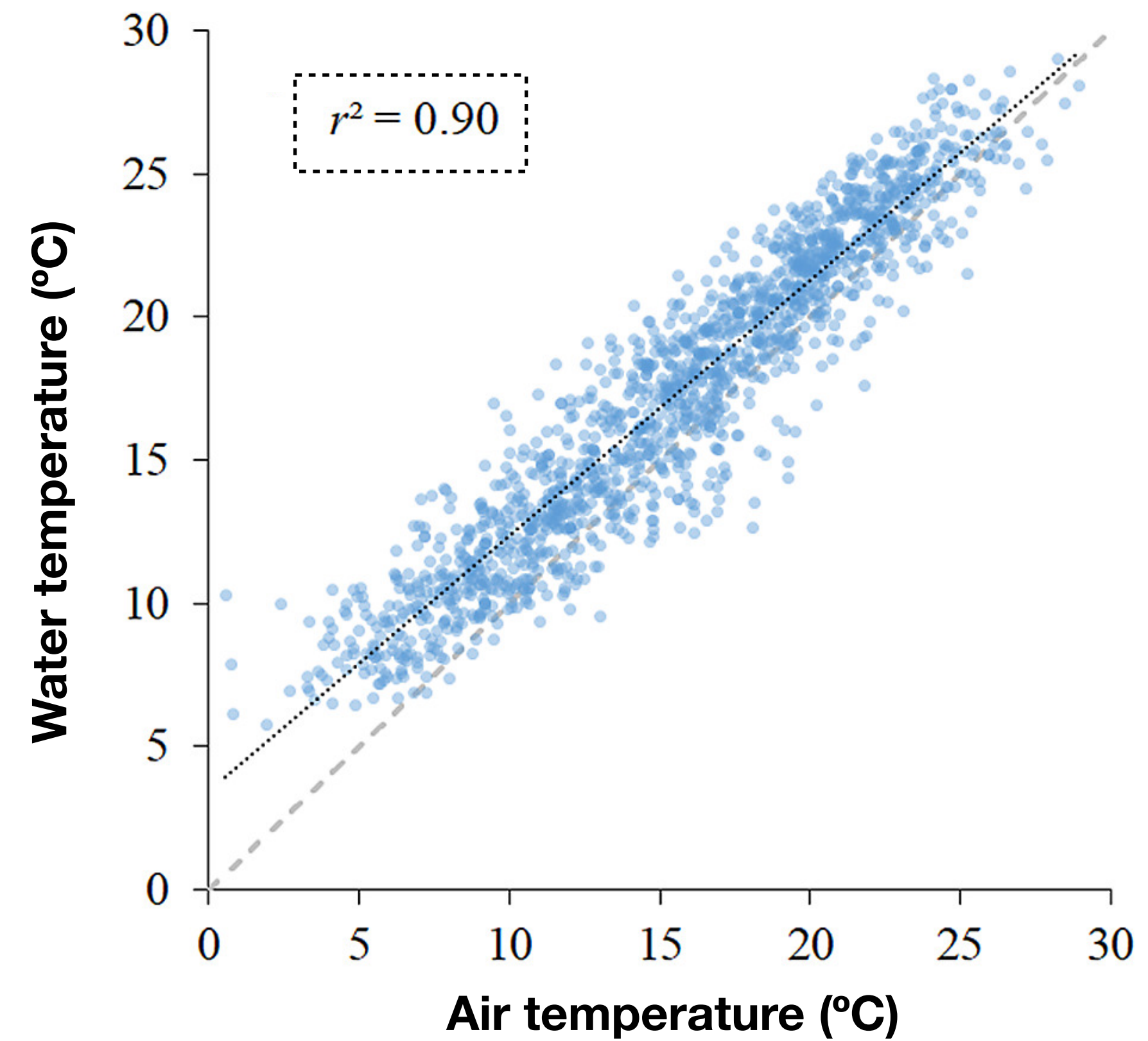
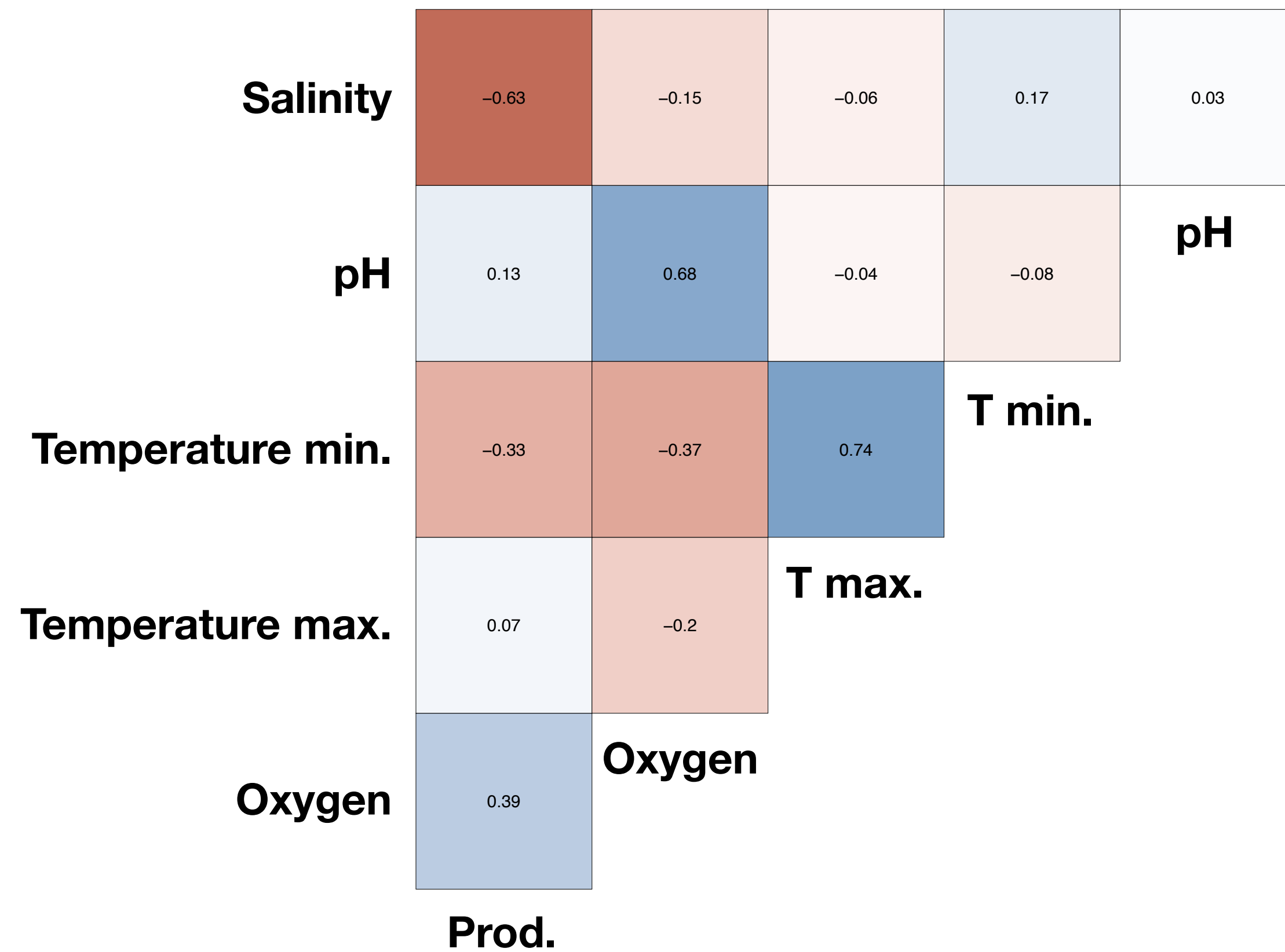
Dissolved molecular oxygen: physiological tolerance limits.

Bathymetry: defines depth range.

Topographic variables (slope): Substrate availability, exposure to currents.

(For symbiotic zooxanthellate) PAR, Diffuse attenuation: Light for photosynthesis.

(For azooxanthellate) Primary productivity / Chlorophyll: Proxy for food supply.



Environmental variables: considerations

Many statistical algorithms (e.g., GLM) have problems fitting data if predictors are strongly correlated, impeding proper characterization of the niche. A collinearity test can be made to discard strongly correlated predictors ($R > 0.8$)

Defining the study area

Calibration vs. projection area:

Calibration area: where the model is trained, is build.

Projection area: where the model is applied to make predictions (can be broader, e.g., globally, future climates, new regions for invasion risk).

The **choice of calibration area significantly influences the model**, and therefore performance and transferability.

Defining the study area

Major question while defining the calibration area:

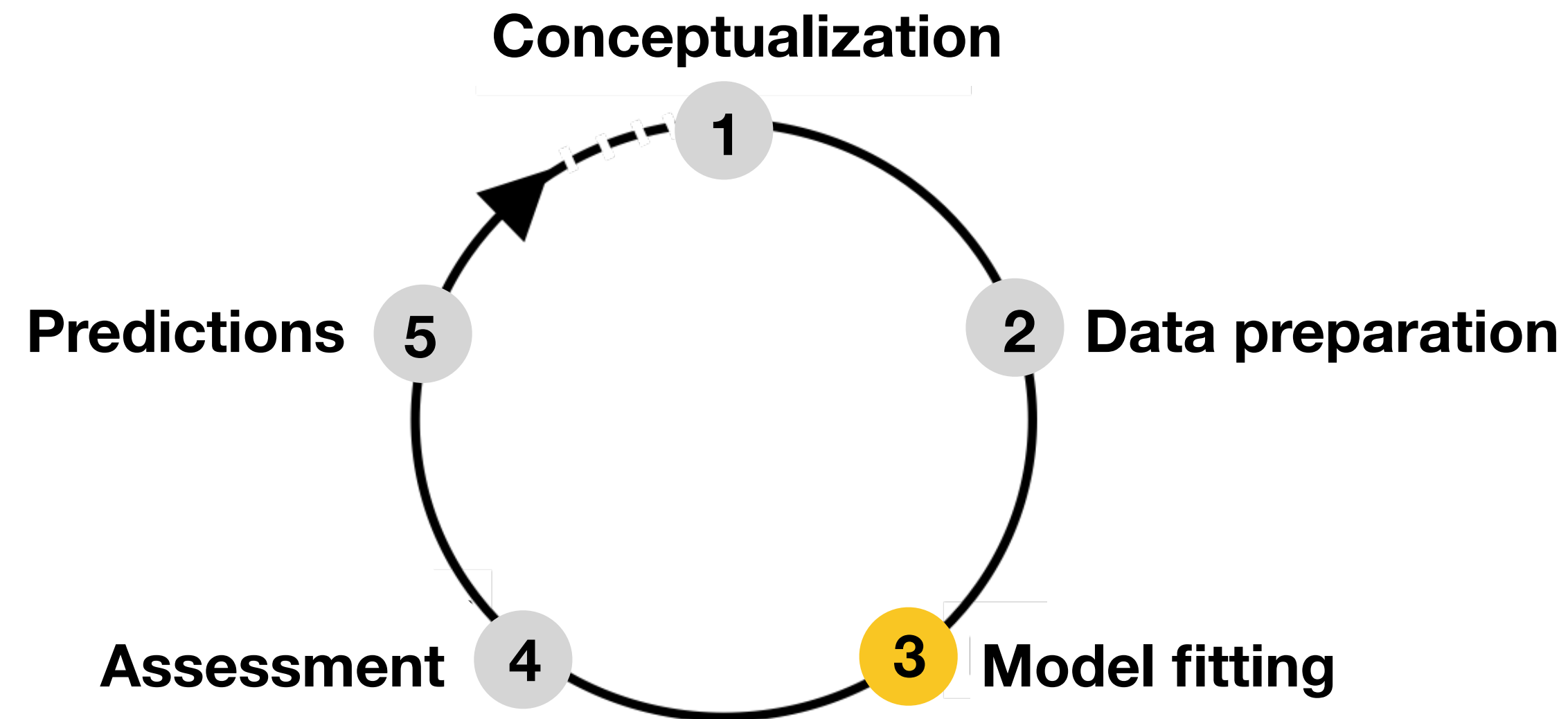
What geographic regions has the species had access to over time? (Considering dispersal ability) - this is the species' accessible area ('M' in BAM):

The calibration area must encompass the accessible region to capture the range of conditions the species experiences.

Rule of Thumb: choose an area relevant to the species' dispersal ecology, large enough to capture the environmental gradients. Avoid environmentally truncated areas (e.g., sampling only the warm edge of a species).

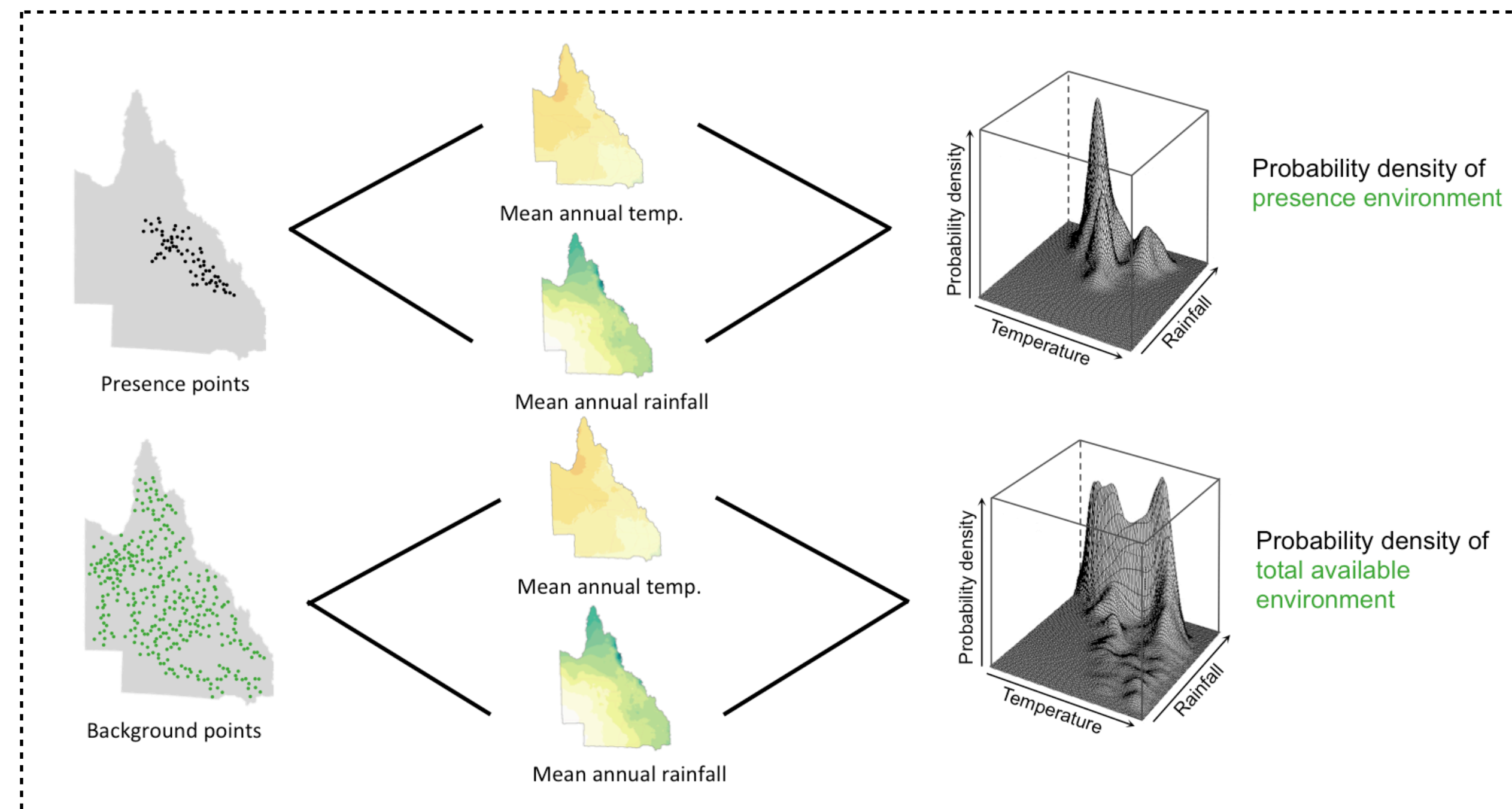
e.g.,

The dispersal potential of a coral species is of 10 days; in the ocean this can be translated onto 300 km of maximum dispersal potential. The study area can be defined as the buffer of 300km around any observed record.



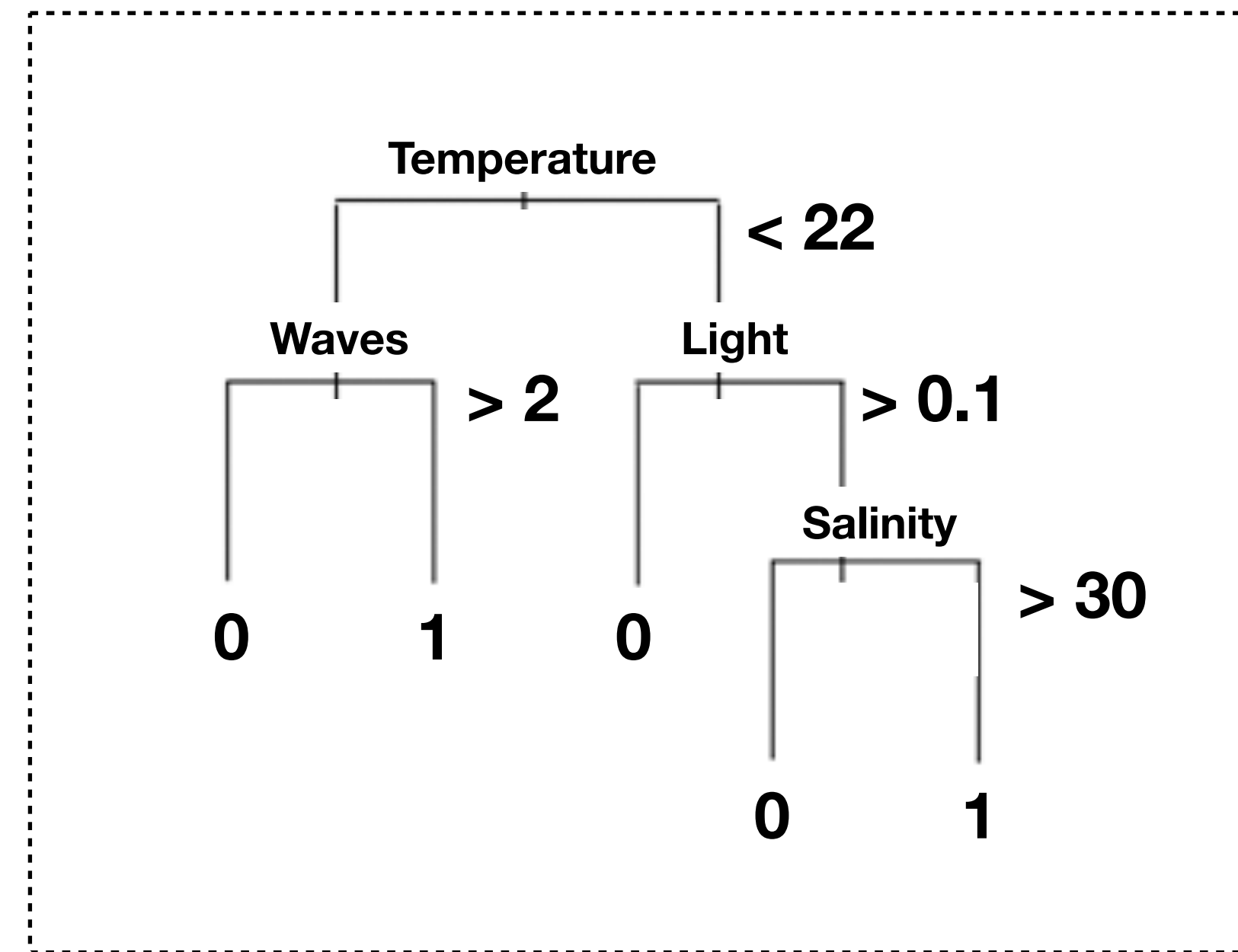
Model fitting

Algorithm selection: choose appropriate statistical method and the algorithm-specific parameters (e.g., regularization in MaxEnt, tree complexity in BRT).



Maximum Entropy predicts occurrences by **comparing the density in the environmental conditions at the locations where the species has been found, with the environmental conditions across the study region.**

MaxEnt **outputs the environmental suitability of a species in the study area** (i.e., the potential distribution).

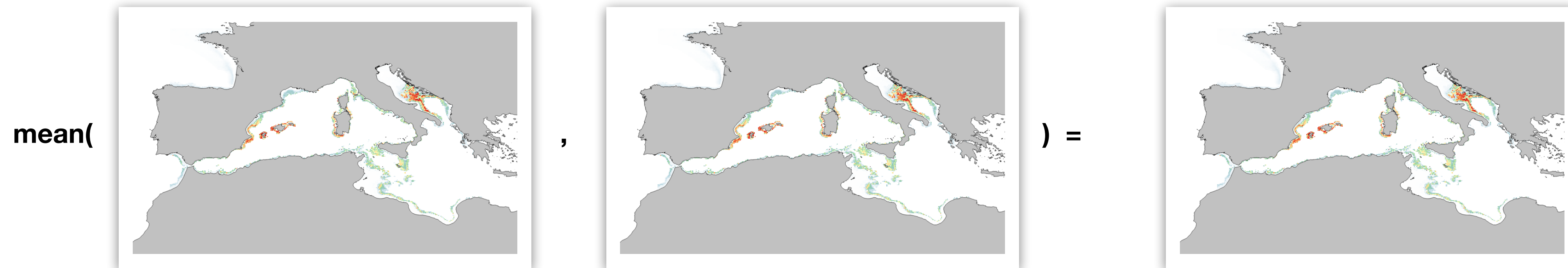


Boosted Regression Trees is a flexible and high performance algorithm, which combine the strengths of **decision trees** (models that relate a response to their predictors by recursive binary splits) **and boosting** (combines simple models to give improved predictive performance).

BRT gives the probability of a species occurring in the study area.

Ensembling models

Instead of using a single-model to investigate species distributions, one can conduct an **ensemble experiment**. This **reduces the uncertainty of algorithms by combining their outputs with a general statistics** (e.g., mean).



One can synthesize the results of multiple algorithms in a single map.

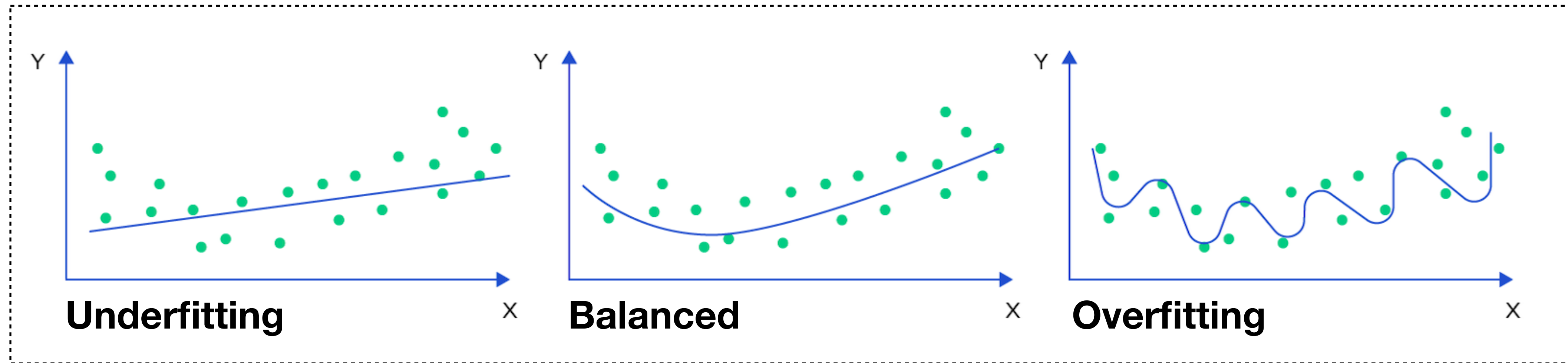
This approach **allows to identify regions of agreement / disagreement between algorithms** (e.g., variance or standard deviation)

Desired properties of model fitting

Deductive and general: develop hypotheses around the causes of the pattern explained by the model, aiming at achieving broad ecological conclusions.

Simple: highlight a few mechanisms without becoming entangled in complex interactions between many variables.

Parsimonious: preferring the simpler of two equally adequate models; favor simple explanations over complex ones.

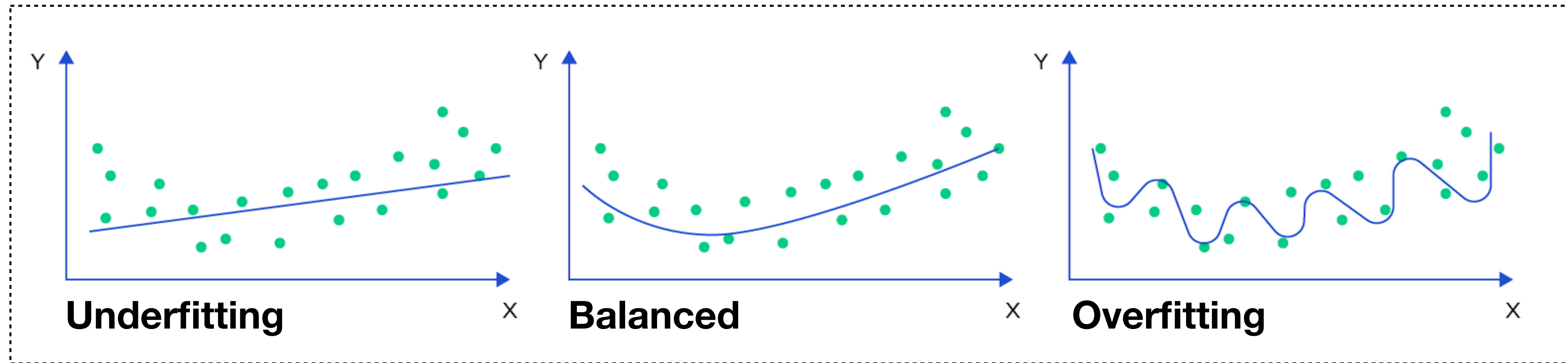


Model fitting pitfalls: underfitting

The model is **too simple to capture the underlying patterns in the data**.

Sources: simplistic algorithms (e.g., a linear model for non-linear relationships). Few or irrelevant variables. Insufficient occurrence data to represent the species' niche. Applying excessive regularization during model fitting.

Issues: Poor predictive performance. Species-environment relationships are missed, leading to unrealistic outputs.

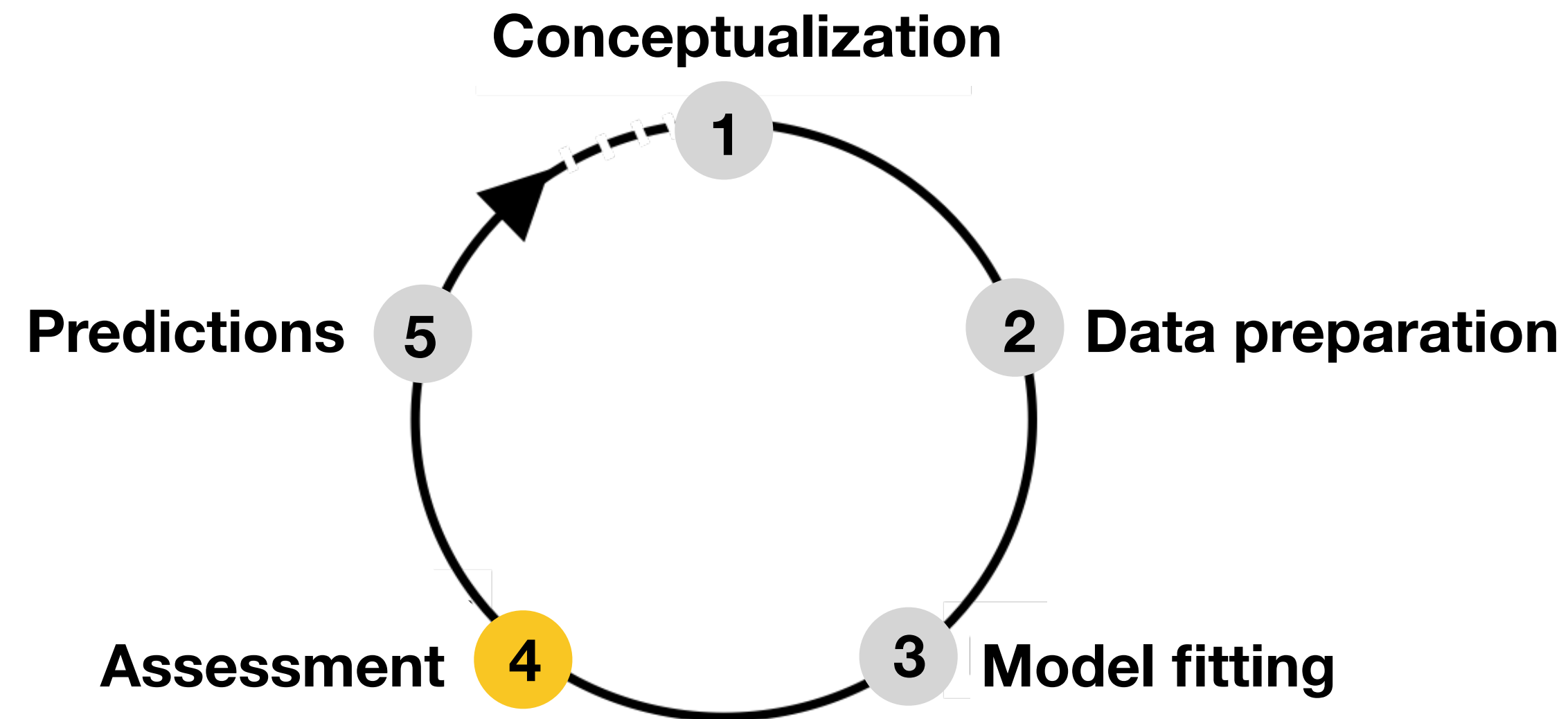


Model pitfalls: overfitting

The model learns the training data too well, including random noise and very specific patterns of the sample, rather than the general patterns.

Sources: overly flexible model algorithms. Including too many environmental variables. Not applying appropriate regularization during model training.

Issues: performs poorly on new (test) data. Creates complex, biologically implausible distribution maps (e.g., suitability in tiny, isolated environmental pockets based on noise).



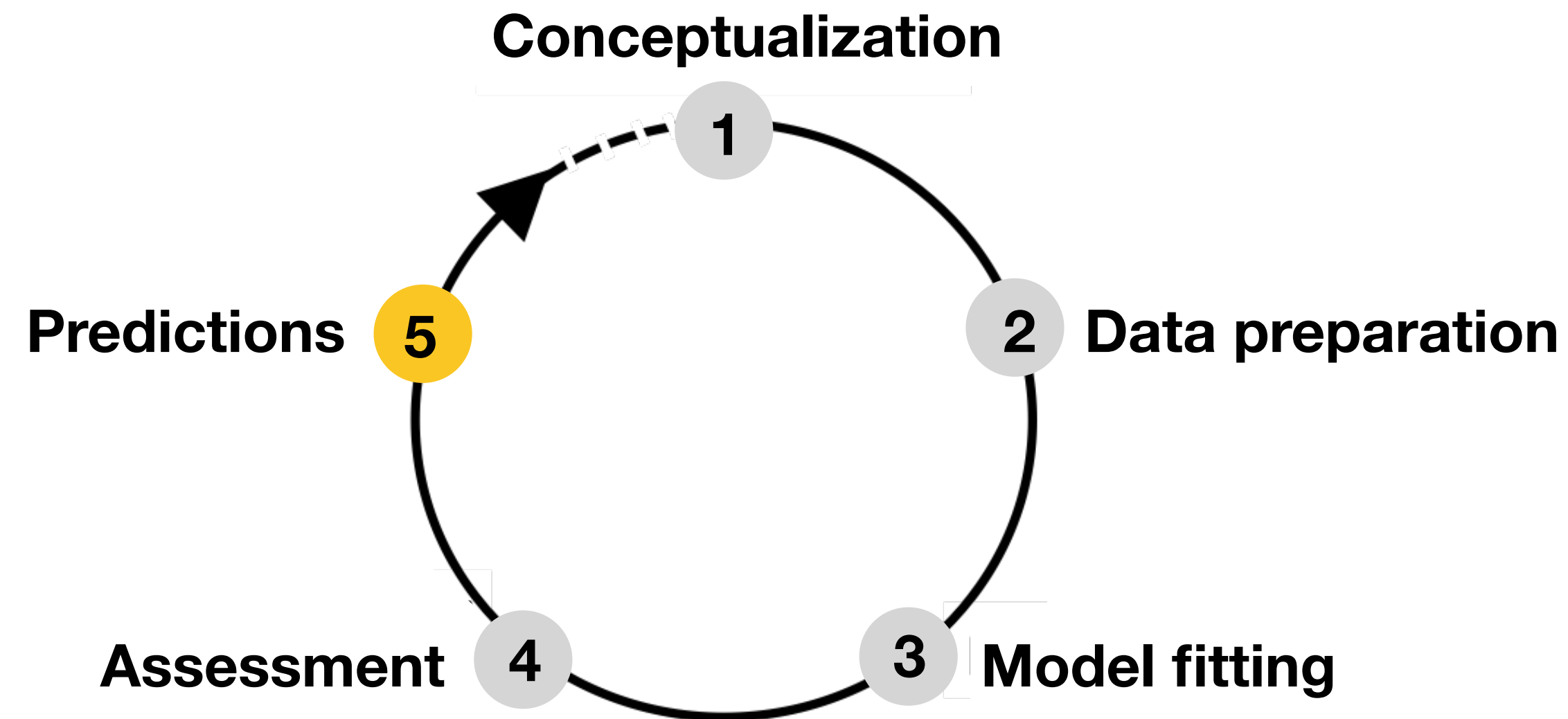
Assessment of the model

How good is the model?

Evaluating performance, as well as ecological realism of fitted response functions.

Does the model accurately distinguish presence sites?

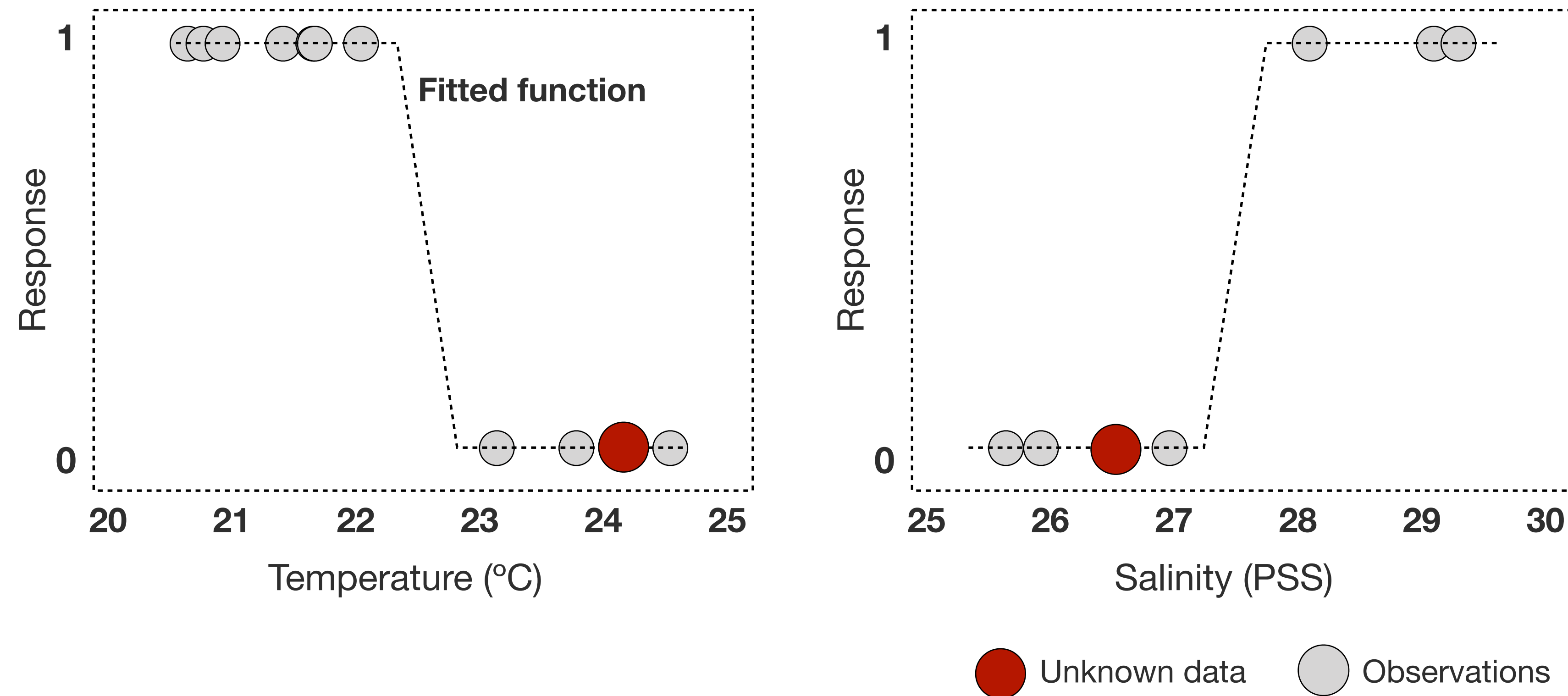
Do the modeled relationships make biological sense? (ecological realism check)



Predictions in space and time

Spatial projection: apply the fitted model to environmental data across the study area to map species' probability of occurrence.

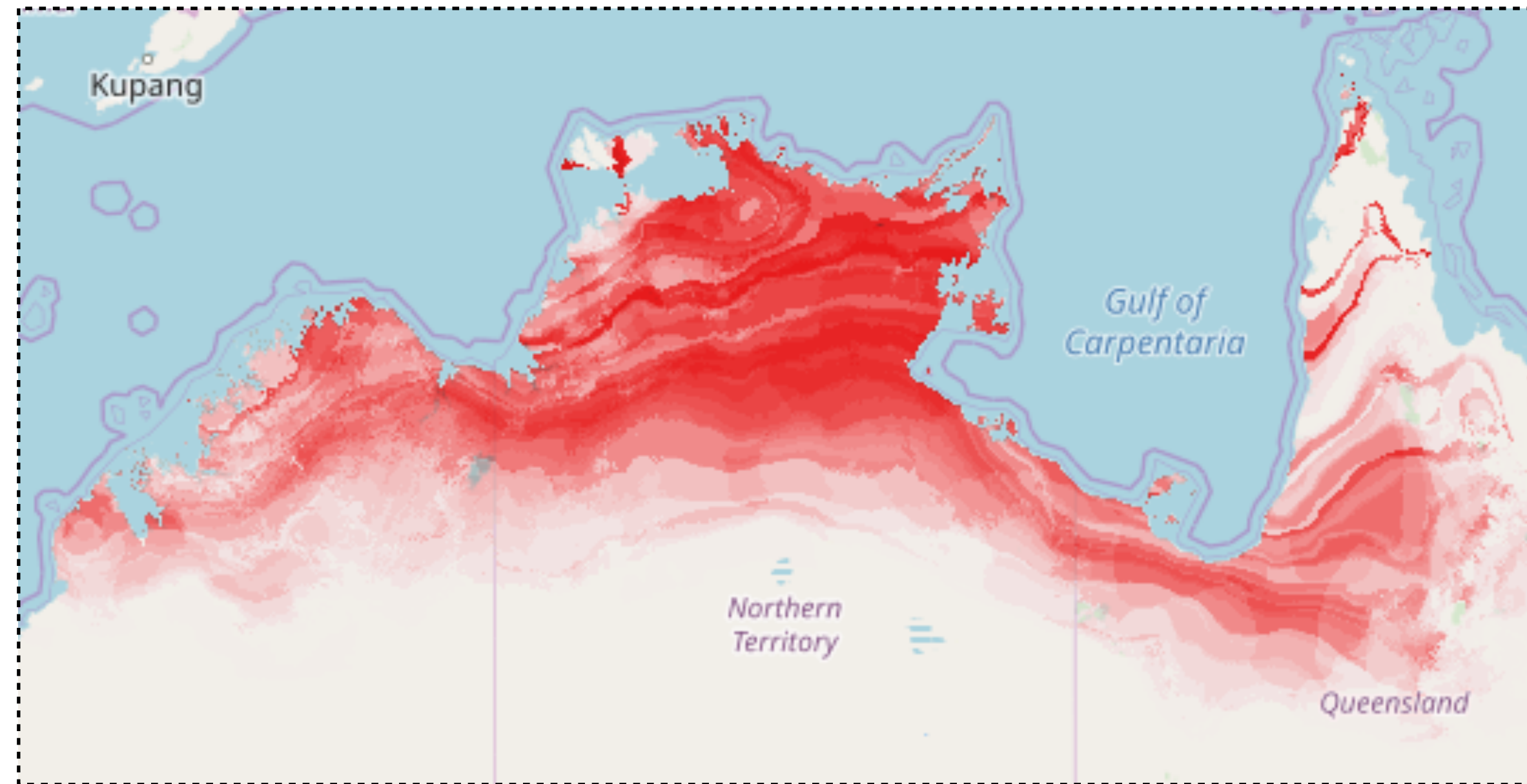
Application: use predictions to address the initial conceived objective (e.g., identify conservation priorities, map future range shifts, assess invasion risk).



Model prediction

If a correlative model can explain the relationship between distribution records and environmental variables, we can make predictions to unknown samples.

e.g., Temp. = 24.5°C & Sal. = 26.5 PSS, response is 0 (i.e., absence).

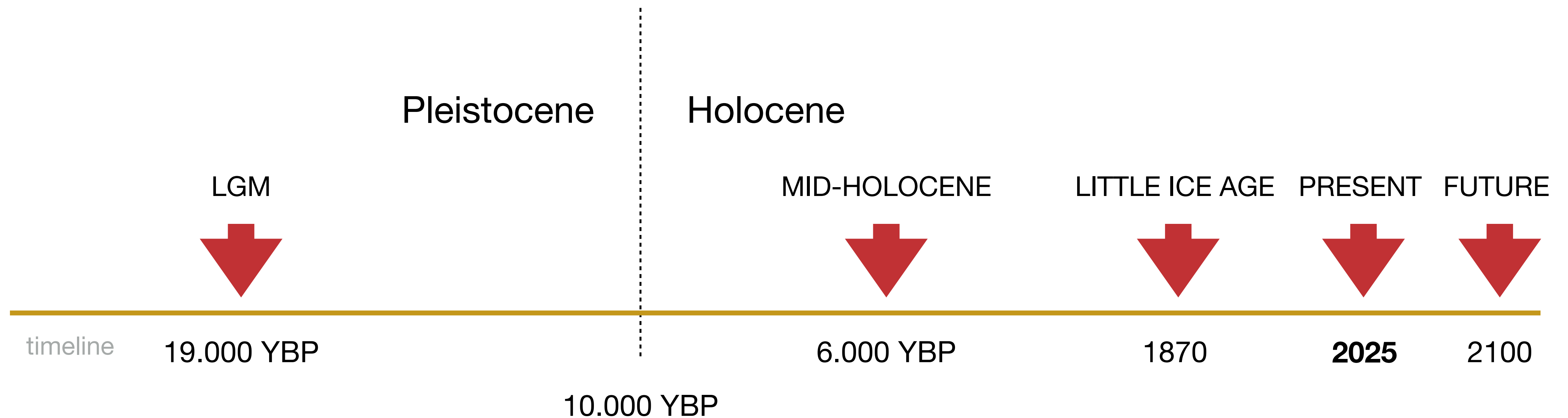


Predictive model-based interpolation

Predictions to **new sites within the range of values of environmental conditions sampled in the training data.**

From **scattered presence records to continuous distribution surfaces.**

Maps do not show where species occur, but rather the distribution of regions with suitable environmental conditions.

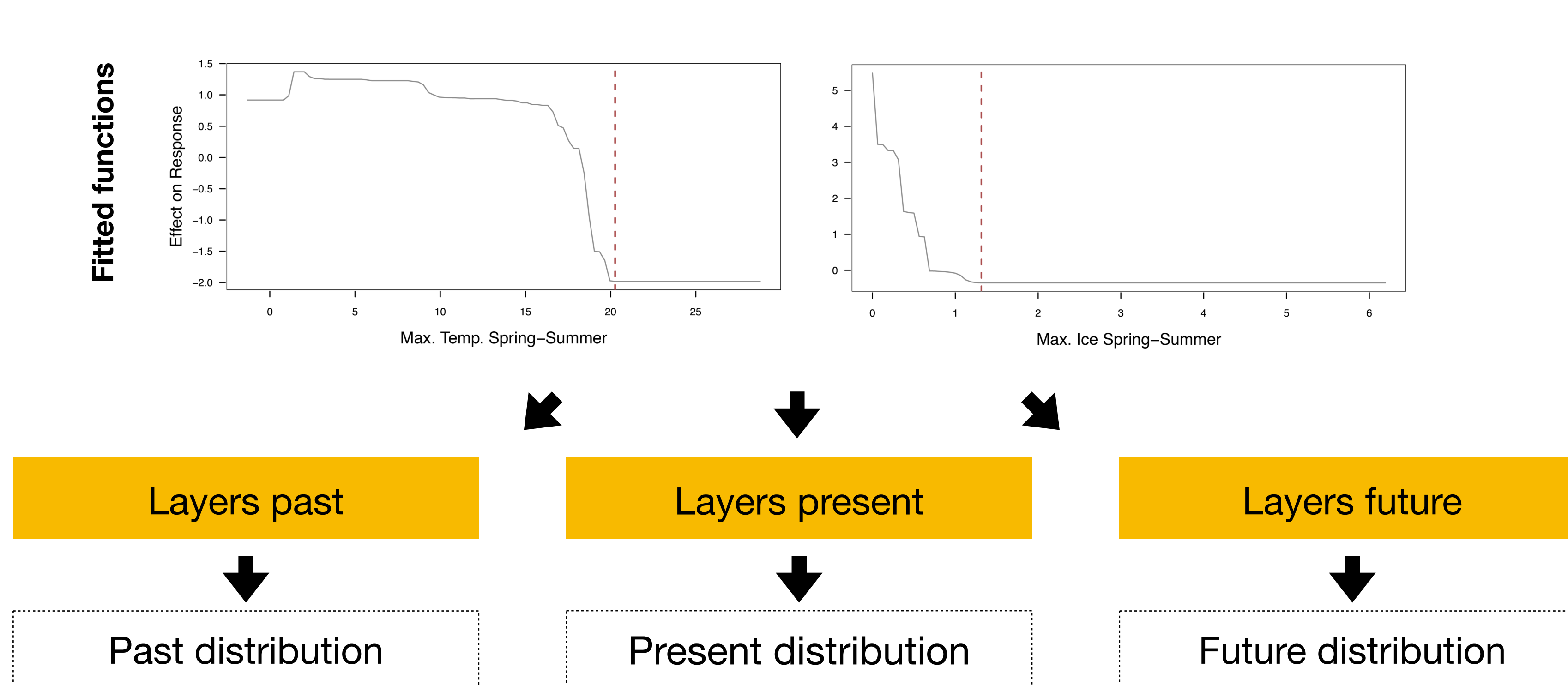


Predictive model-based transferability

Ability of a model to predict beyond the data it was originally trained on.

Process: a model fits **observed species occurrence data against environmental variables for the corresponding period**. The **same model is then applied directly to new climate data** representing new environmental conditions.

Predictive model-based transferability

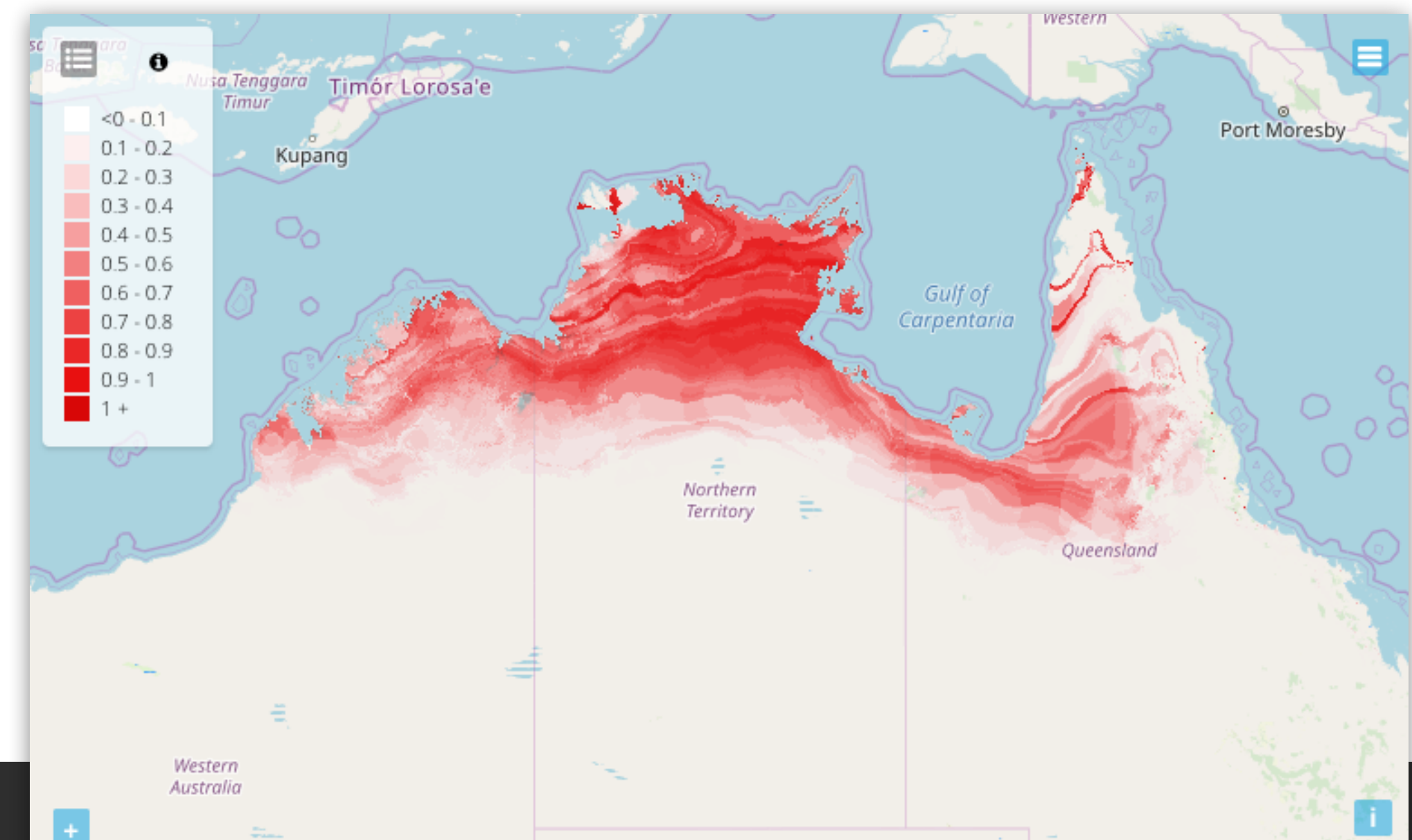
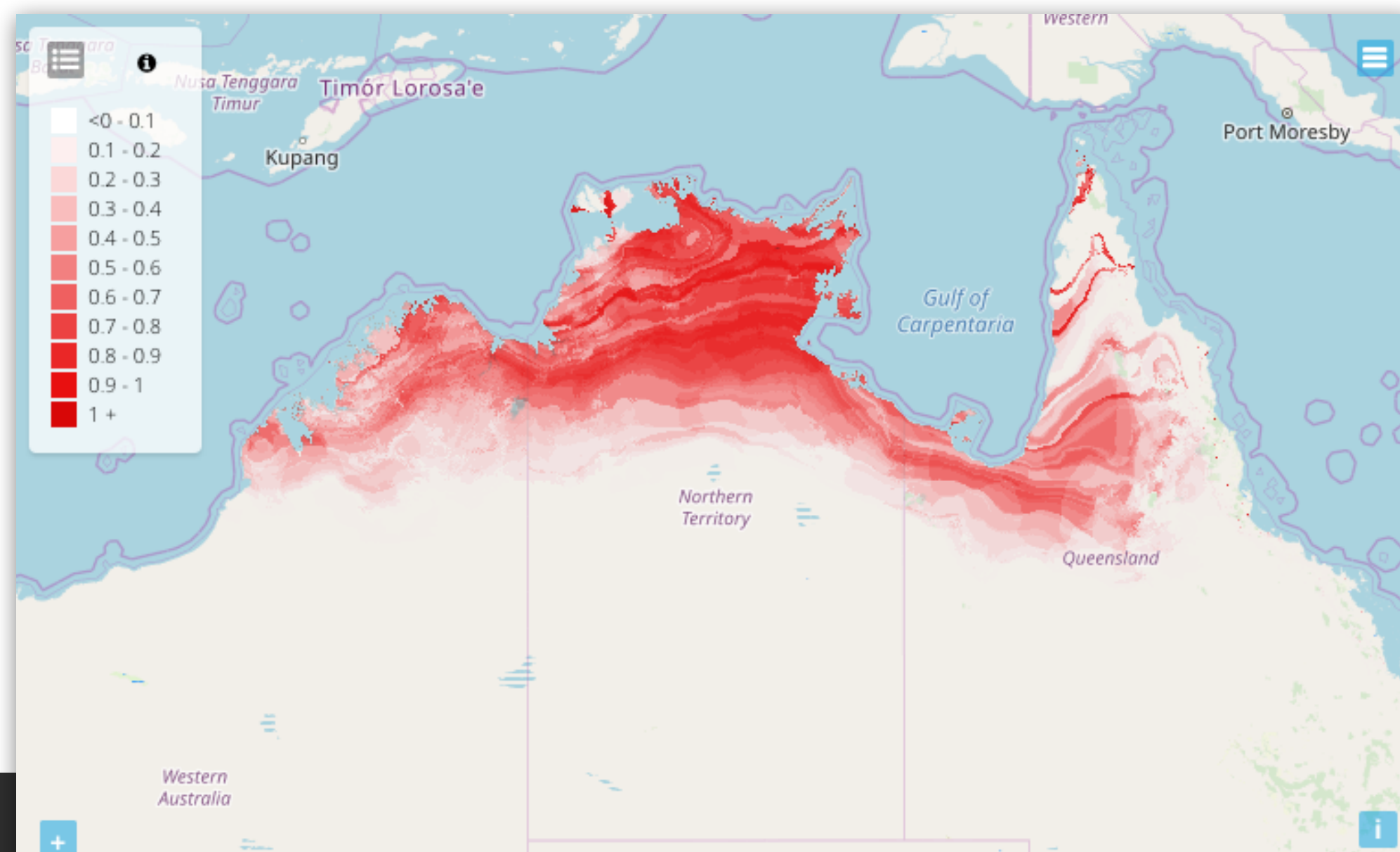


Regardless of the time period / region, **all environmental layers must be included in the transferability process**. So, the **availability of layers for climate scenarios also determines the choice of predictors for model fitting**.

Predictive model-based transferability

Model transferability can be analyzed (per cell) with different approaches:

(1) comparing **predictions** (e.g., probability) under different conditions.

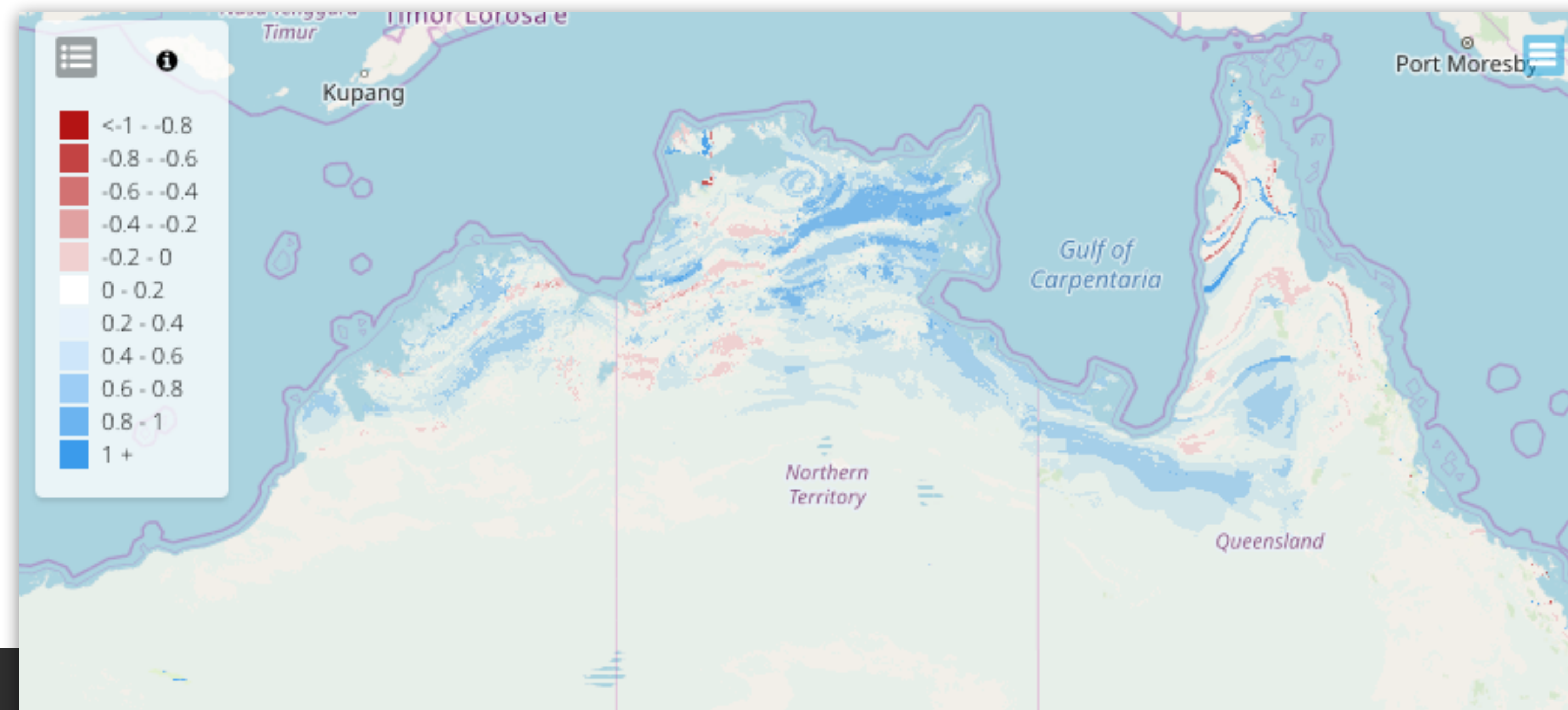


(2) **change in probability**, i.e., the difference in the predicted probabilities of different conditions;

e.g.,

[Layer 1 Future probability] - [Layer 2 present-day probability]

The map scales from -1 to 1, where negative values refer to lower suitable conditions and positive higher suitability.



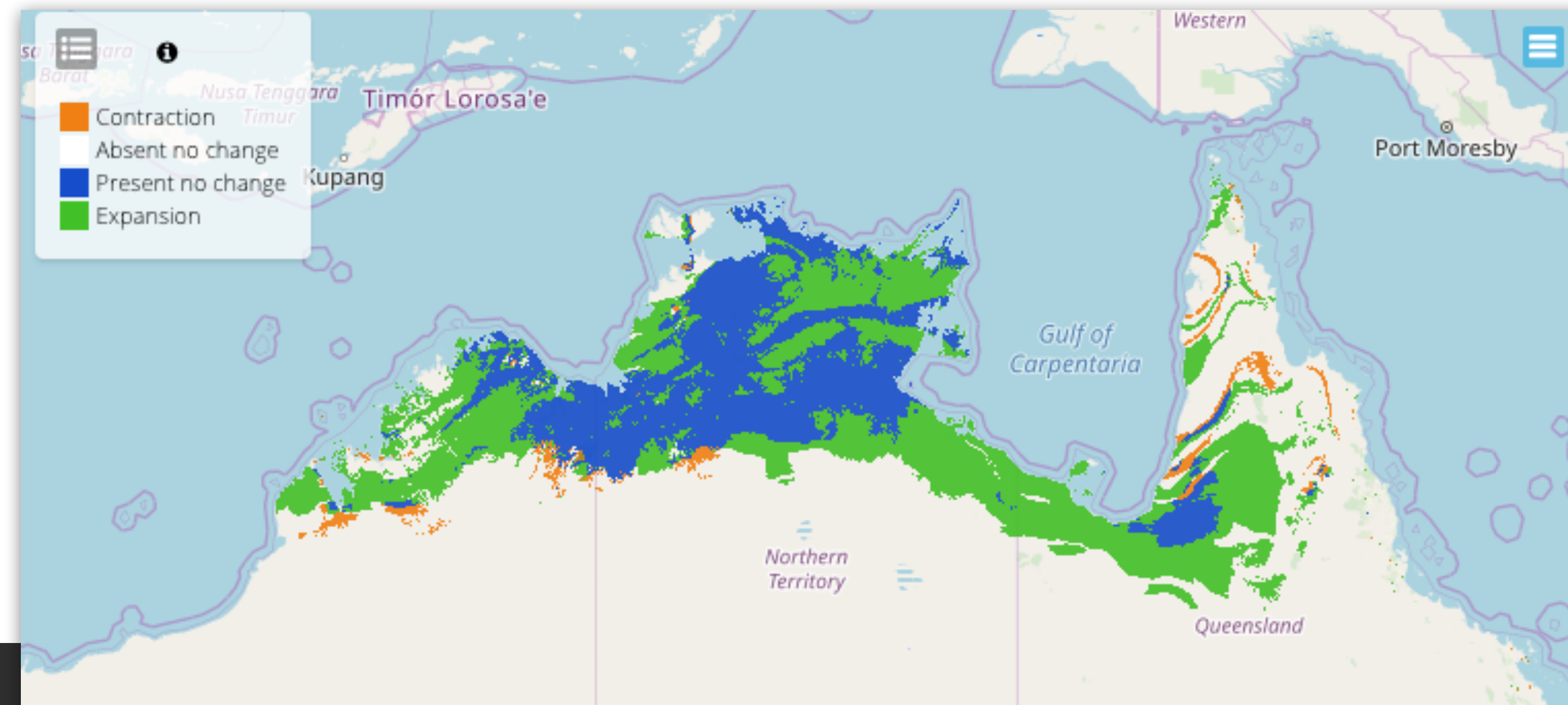
(3) **change in species distributional range**, generated with binary maps (i.e., 0 or 1).

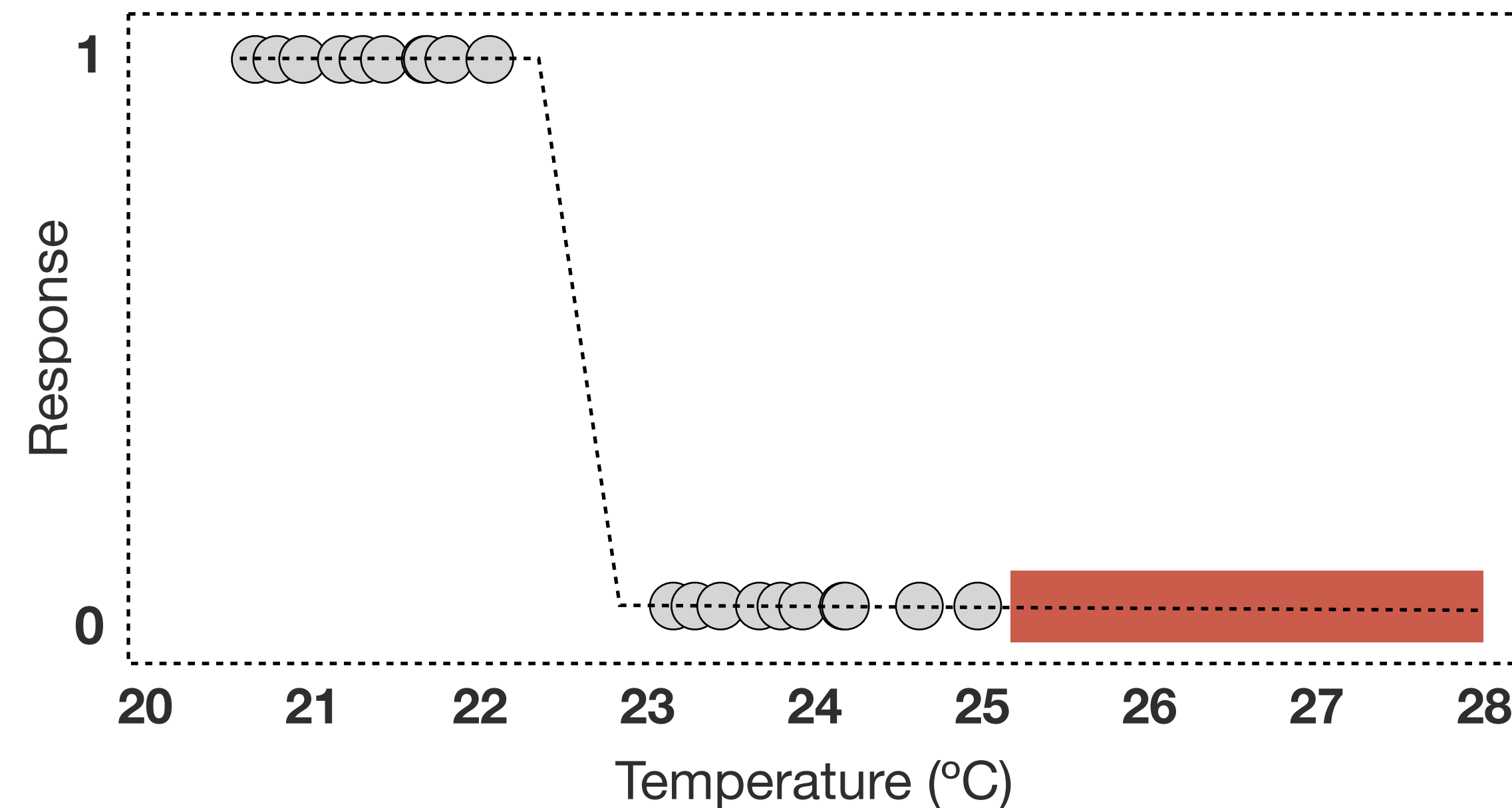
Comparing maps can indicate:

No change: present distribution equals future distribution;

Range loss or contraction: presence in the present and absence in future;

Range expansion: absence in the present and presence in the future.





Model transferability may lead to extrapolation, i.e., predictions to environmental values outside the range of environmental values used to fit the model.

e.g.,

A model fitted records with temperatures of 20-25°C. If predictions are made for temperatures > 25°C, then the model will extrapolate. No information exists on the probability of occurrence at > 25°C, so the predictions will be uncertain.

Core assumptions of SDM

Equilibrium: species are in equilibrium with the environmental conditions - species occur in all suitable areas and are absent from all unsuitable areas.

Niche representation: sampled occurrences adequately represent the species' environmental tolerances.

Full predictor set: environmental variables influencing the distribution are considered.

Unlimited dispersal (often implied): dispersal does not determine the occurrence of species within suitable environments.

Limited biotic influence (often implied): biotic interactions (e.g., competition, predation) don't determine the occurrence of species within suitable environments.

*** These assumptions are often violated to some degree.*

References

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